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| **1** | **Feature Transformation (Any one)**<br><br>**A. To use PCA Algorithm for dimensionality reduction.**<br><br>**You have a dataset that includes measurements for different variables on wine (alcohol, ash, magnesium, and so on). Apply PCA algorithm & transform this data so that most variations in the measurements of the variables are captured by a small number of principal components so that it is easier to distinguish between red and white wine by inspecting these principal components.**<br><br>**Dataset Link: https://media.geeksforgeeks.org/wp-content/uploads/Wine.csv**<br><br>**B. Apply LDA Algorithm on Iris Dataset and classify which species a given flower belongs to.**<br><br>**Dataset Link:https://www.kaggle.com/datasets/uciml/iris** |
| **2** | **Regression Analysis:(Any one)**<br><br>**A. Predict the price of the Uber ride from a given pickup point to the agreed drop-off location. Perform following tasks:**<br><br>**1. Pre-process the dataset.**<br><br>**2. Identify outliers.**<br><br>**3. Check the correlation.**<br><br>**4. Implement linear regression and ridge, Lasso regression models.**<br><br>**5. Evaluate the models and compare their respective scores like R2, RMSE, etc.**<br><br>**Dataset link: https://www.kaggle.com/datasets/yasserh/uber-fares-dataset**<br><br>**B. Use the diabetes data set from UCI and Pima Indians Diabetes data set for performing the following:**<br><br>**a. Univariate analysis: Frequency, Mean, Median, Mode, Variance, Standard Deviation, Skewness and Kurtosis**<br><br>**b. Bivariate analysis: Linear and logistic regression modeling**<br><br>**c. Multiple Regression analysis**<br><br>**d. Also compare the results of the above analysis for the two data sets**<br><br>**Dataset link: https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database** |

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| **3** | **Classification Analysis (Any one)**<br>**A. Implementation of Support Vector Machines (SVM) for classifying images of hand-written digits into their respective numerical classes (0 to 9).**<br>**B. Implement K-Nearest Neighbours' algorithm on Social network ad dataset. Compute confusion matrix, accuracy, error rate, precision and recall on the given dataset.**<br><br>**Dataset link:https://www.kaggle.com/datasets/rakeshrau/social-network-ads** |
| **4** | **Clustering Analysis (Any one)**<br>**A. Implement K-Means clustering on Iris.csv dataset. Determine the number of clusters using the elbow method.**<br><br>**Dataset Link: https://www.kaggle.com/datasets/uciml/iris**<br>**B. Implement K-Mediod Algorithm on a credit card dataset. Determine the number of clusters using the Silhouette Method.**<br><br>**Dataset link: https://www.kaggle.com/datasets/arjunbhasin2013/ccdata** |
| **5** | **Ensemble Learning (Any one)**<br>**A. Implement Random Forest Classifier model to predict the safety of the car.**<br><br>**Dataset link: https://www.kaggle.com/datasets/elikplim/car-evaluation-data-set**<br>**B. Use different voting mechanism and Apply AdaBoost (Adaptive Boosting), Gradient Tree Boosting (GBM), XGBoost classification on Iris dataset and compare the performance of three models using different evaluation measures.**<br><br>**Dataset Link: https://www.kaggle.com/datasets/uciml/iris** |
| **6** | **Reinforcement Learning (Any one)**<br>**A. Implement Reinforcement Learning using an example of a maze environment that the agent needs to explore.**<br>**B. Solve the Taxi problem using reinforcement learning where the agent acts as a taxi driver to pick up a passenger at one location and then drop the passenger off at their destination.**<br>**C. Build a Tic-Tac-Toe game using reinforcement learning in Python by using following tasks**<br>**a. Setting up the environment**<br>**b. Defining the Tic-Tac-Toe game**<br>**c. Building the reinforcement learning model**<br>**d. Training the model**<br>**e. Testing the model** |

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| **7** | **Data Loading, Storage and File Formats**<br>**Problem Statement: Analyzing Sales Data from Multiple File Formats**<br>**Dataset: Sales data in multiple file formats (e.g., CSV, Excel, JSON)**<br>**Description: The goal is to load and analyze sales data from different file formats, including CSV, Excel, and JSON, and perform data cleaning, transformation, and analysis on the dataset.**<br>**Tasks to Perform:**<br>**Obtain sales data files in various formats, such as CSV, Excel, and JSON.**<br>**1. Load the sales data from each file format into the appropriate data structures or dataframes.**<br>**2. Explore the structure and content of the loaded data, identifying any inconsistencies, missing values, or data quality issues.**<br>**3. Perform data cleaning operations, such as handling missing values, removing duplicates, or correcting inconsistencies.**<br>**4. Convert the data into a unified format, such as a common dataframe or data structure, to enable seamless analysis.**<br>**5. Perform data transformation tasks, such as merging multiple datasets, splitting columns, or deriving new variables.**<br>**6. Analyze the sales data by performing descriptive statistics, aggregating data by specific variables, or calculating metrics such as total sales, average order value, or product category distribution.**<br>**7. Create visualizations, such as bar plots, pie charts, or box plots, to represent the sales data and gain insights into sales trends, customer behavior, or product performance.** |
| **8** | **Interacting with Web APIs**<br>**Problem Statement: Analyzing Weather Data from OpenWeatherMap API**<br>**Dataset: Weather data retrieved from OpenWeatherMap API**<br>**Description: The goal is to interact with the OpenWeatherMap API to retrieve weather data for a specific location and perform data modeling and visualization to analyze weather patterns over time.**<br>**Tasks to Perform:**<br>**1. Register and obtain API key from OpenWeatherMap.**<br>**2. Interact with the OpenWeatherMap API using the API key to retrieve weather data for a specific location.**<br>**3. Extract relevant weather attributes such as temperature, humidity, wind speed, and precipitation from the API response.**<br>**4. Clean and preprocess the retrieved data, handling missing values or inconsistent formats.**<br>**5. Perform data modeling to analyze weather patterns, such as calculating average temperature, maximum/minimum values, or trends over time.**<br>**6. Visualize the weather data using appropriate plots, such as line charts, bar plots, or scatter plots, to represent temperature changes, precipitation levels, or wind speed variations.**<br>**7. Apply data aggregation techniques to summarize weather statistics by specific time periods (e.g., daily, monthly, seasonal).**<br>**8. Incorporate geographical information, if available, to create maps or geospatial visualizations representing weather patterns across different locations.**<br>**9. Explore and visualize relationships between weather attributes, such as temperature and humidity, using correlation plots or heatmaps.** |

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| **9** | **Data Cleaning and Preparation**<br>**Problem Statement: Analyzing Customer Churn in a Telecommunications Company**<br>**Dataset: "Telecom_Customer_Churn.csv"**<br>**Description: The dataset contains information about customers of a telecommunications company and whether they have churned (i.e., discontinued their services). The dataset includes various attributes of the customers, such as their demographics, usage patterns, and account information. The goal is to perform data cleaning and preparation to gain insights into the factors that contribute to customer churn.**<br>**Tasks to Perform:**<br>**1. Import the "Telecom_Customer_Churn.csv" dataset.**<br>**2. Explore the dataset to understand its structure and content.**<br>**3. Handle missing values in the dataset, deciding on an appropriate strategy.**<br>**4. Remove any duplicate records from the dataset.**<br>**5. Check for inconsistent data, such as inconsistent formatting or spelling variations, and standardize it.**<br>**6. Convert columns to the correct data types as needed.**<br>**7. Identify and handle outliers in the data.**<br>**8. Perform feature engineering, creating new features that may be relevant to predicting customer churn.**<br>**9. Normalize or scale the data if necessary.**<br>**10. Split the dataset into training and testing sets for further analysis.**<br>**11. Export the cleaned dataset for future analysis or modeling.** |
| **10** | **Data Wrangling**<br>**Problem Statement: Data Wrangling on Real Estate Market**<br>**Dataset: "RealEstate_Prices.csv"**<br>**Description: The dataset contains information about housing prices in a specific real estate market. It includes various attributes such as property characteristics, location, sale prices, and other relevant features. The goal is to perform data wrangling to gain insights into the factors influencing housing prices and prepare the dataset for further analysis or modeling.**<br>**Tasks to Perform:**<br>**1. Import the "RealEstate_Prices.csv" dataset. Clean column names by removing spaces, special characters, or renaming them for clarity.**<br>**2. Handle missing values in the dataset, deciding on an appropriate strategy (e.g., imputation or removal).**<br>**3. Perform data merging if additional datasets with relevant information are available (e.g., neighborhood demographics or nearby amenities).**<br>**4. Filter and subset the data based on specific criteria, such as a particular time period, property type, or location.**<br>**5. Handle categorical variables by encoding them appropriately (e.g., one-hot encoding or label encoding) for further analysis.**<br>**6. Aggregate the data to calculate summary statistics or derived metrics such as average sale prices by neighborhood or property type.**<br>**7. Identify and handle outliers or extreme values in the data that may affect the analysis or modeling process.** |

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| **11** | **Data Visualization using matplotlib**<br>**Problem Statement: Analyzing Air Quality Index (AQI) Trends in a City**<br>**Dataset: "City_Air_Quality.csv"**<br>**Description: The dataset contains information about air quality measurements in a specific city over a period of time. It includes attributes such as date, time, pollutant levels (e.g., PM2.5, PM10, CO), and the Air Quality Index (AQI) values. The goal is to use the matplotlib library to create visualizations that effectively represent the AQI trends and patterns for different pollutants in the city.**<br>**Tasks to Perform:**<br>**1. Import the "City_Air_Quality.csv" dataset.**<br>**2. Explore the dataset to understand its structure and content.**<br>**3. Identify the relevant variables for visualizing AQI trends, such as date, pollutant levels, and AQI values.**<br>**4. Create line plots or time series plots to visualize the overall AQI trend over time.**<br>**5. Plot individual pollutant levels (e.g., PM2.5, PM10, CO) on separate line plots to visualize their trends over time.**<br>**6. Use bar plots or stacked bar plots to compare the AQI values across different dates or time periods.**<br>**7. Create box plots or violin plots to analyze the distribution of AQI values for different pollutant categories.**<br>**8. Use scatter plots or bubble charts to explore the relationship between AQI values and pollutant levels.**<br>**9. Customize the visualizations by adding labels, titles, legends, and appropriate color schemes.** |
| **12** | **Data Aggregation**<br>**Problem Statement: Analyzing Sales Performance by Region in a Retail Company**<br>**Dataset: "Retail_Sales_Data.csv"**<br>**Description: The dataset contains information about sales transactions in a retail company. It includes attributes such as transaction date, product category, quantity sold, and sales amount. The goal is to perform data aggregation to analyze the sales performance by region and identify the top-performing regions.**<br>**Tasks to Perform:**<br>**1. Import the "Retail_Sales_Data.csv" dataset.**<br>**2. Explore the dataset to understand its structure and content.**<br>**3. Identify the relevant variables for aggregating sales data, such as region, sales amount, and product category.**<br>**4. Group the sales data by region and calculate the total sales amount for each region.**<br>**5. Create bar plots or pie charts to visualize the sales distribution by region.**<br>**6. Identify the top-performing regions based on the highest sales amount.**<br>**7. Group the sales data by region and product category to calculate the total sales amount for each combination.**<br>**8. Create stacked bar plots or grouped bar plots to compare the sales amounts across different regions and product categories.** |

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| **13** | **Time Series Data Analysis**<br>**Problem statement: Analysis and Visualization of Stock Market Data**<br>**Dataset: "Stock_Prices.csv"**<br>**Description: The dataset contains historical stock price data for a particular company over a period of time. It includes attributes such as date, closing price, volume, and other relevant features. The goal is to perform time series data analysis on the stock price data to identify trends, patterns, and potential predictors, as well as build models to forecast future stock prices.**<br>**Tasks to Perform:**<br>**1. Import the "Stock_Prices.csv" dataset.**<br>**2. Explore the dataset to understand its structure and content.**<br>**3. Ensure that the date column is in the appropriate format (e.g., datetime) for time series analysis.**<br>**4. Plot line charts or time series plots to visualize the historical stock price trends over time.**<br>**5. Calculate and plot moving averages or rolling averages to identify the underlying trends and smooth out noise.**<br>**6. Perform seasonality analysis to identify periodic patterns in the stock prices, such as weekly, monthly, or yearly fluctuations.**<br>**7. Analyze and plot the correlation between the stock prices and other variables, such as trading volume or market indices.**<br>**8. Use autoregressive integrated moving average (ARIMA) models or exponential smoothing models to forecast future stock prices.** |
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